



Knochen- und Prothesen-assoziierte Infektionen Diagnostik und Therapie

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2026

ABS Kurs

Interessenkonflikte (2023 – 2026)

- Lecture fees: AbbVie, Cogitando, GSK, Infectopharm, Medupdate, Pfizer, Phaion, Shionogi, Tillotts
- Travel grants: Pfizer
- Advisory board: MSD, Shionogi
- Editorial board: Die Innere Medizin- Springer Verlag; Podcast –Consilium infectorum – Infectopharm
- Scientific Board: Streamed-up -InfektioLive

Wann starten?

- Patienten stabil – erst Diagnostik!

Welche Erreger erwarten wir?

- Abhängig von der Entität und dem Risikoprofil der PatientInnen

Knocheninfektionen Entitäten

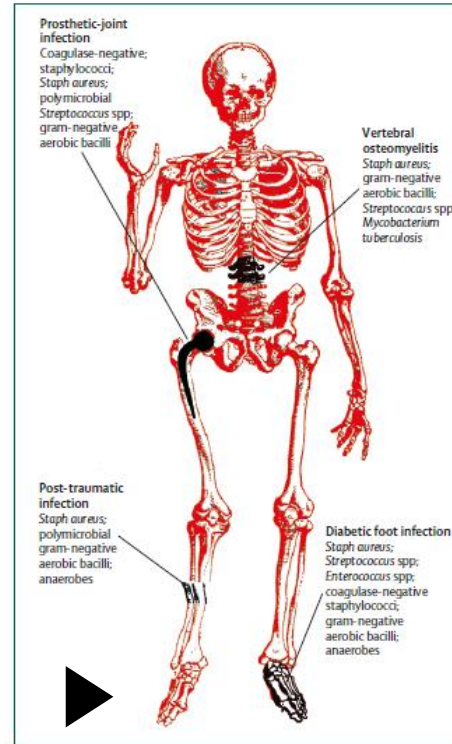
Lee and Waldvogel (2004) Lancet

per-continuitatem

Diabetische Fußinfektion

Periprothetische
Infektion

Posttraumatische Infektion



hämatogen

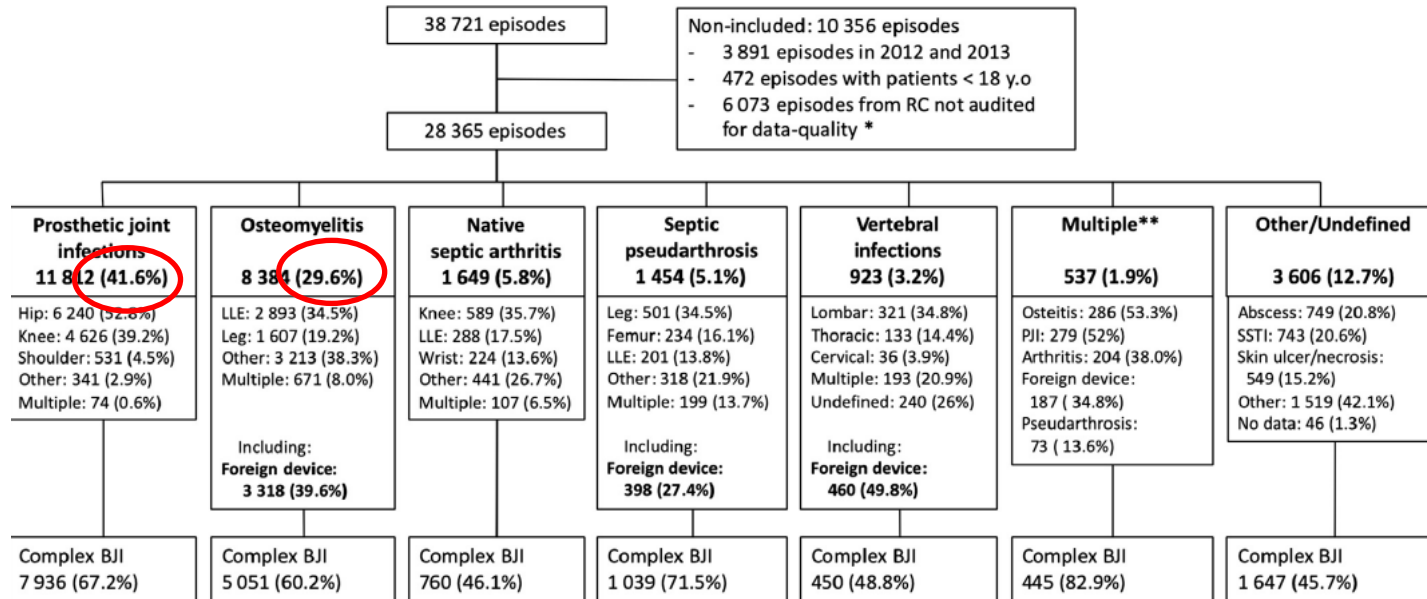
Spondylodiszitis

Osteomyelitis

Septische Arthritis

Epidemiologie – PJI gefolgt von Osteomyelitis

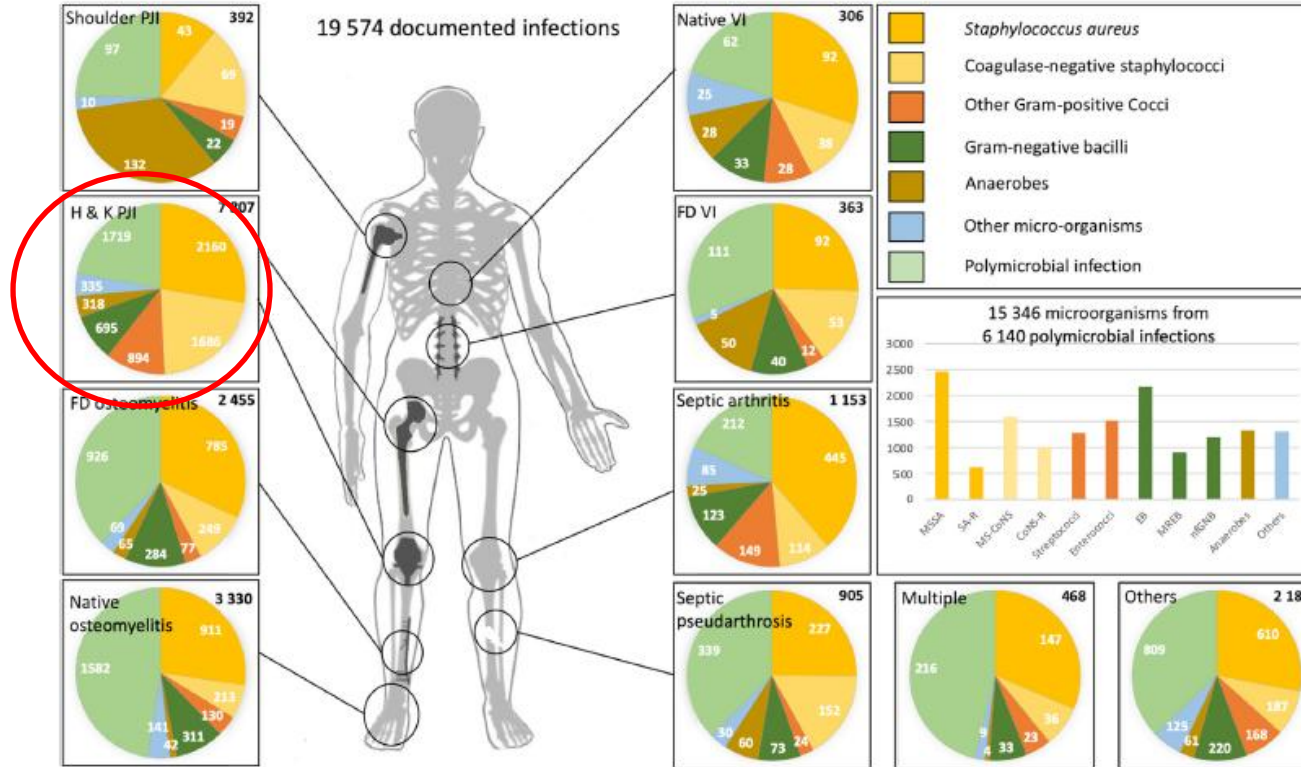
Lamignen et al., (2021) J Infect.



Erreger: *S. aureus* führend – aber bunt Spektrum je nach Lokalisation



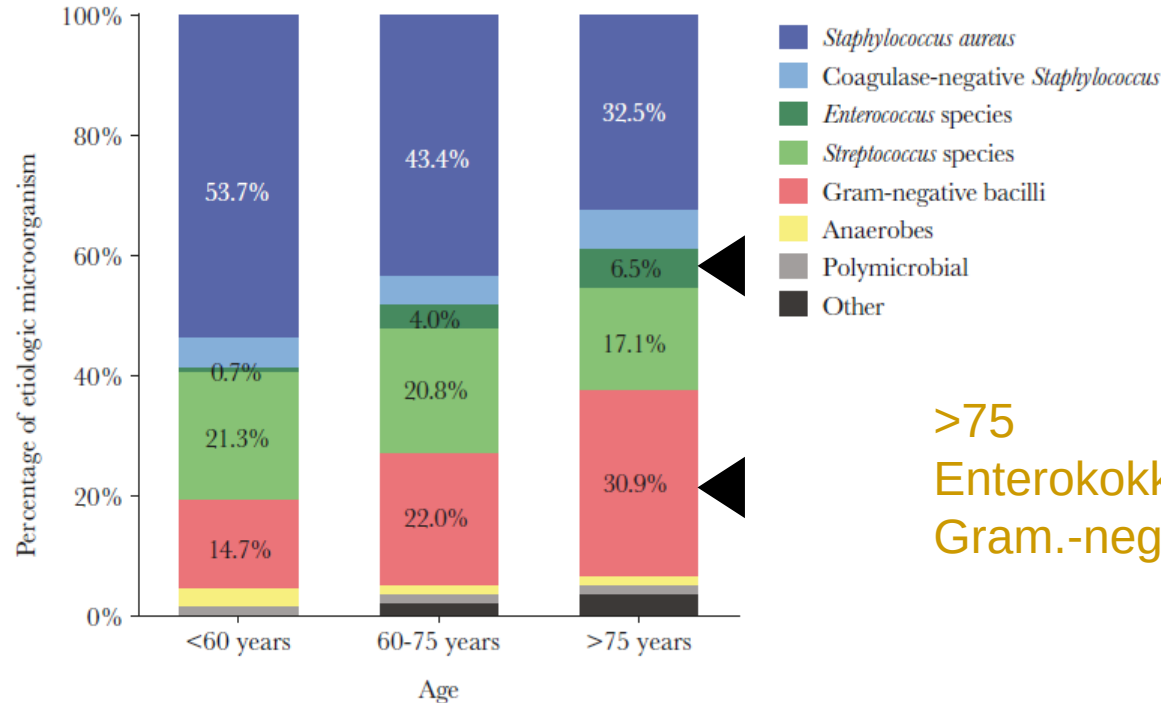
Lamignen et al., (2021) J Infect.





Erreger: abhängig vom Alter

Kim et al., (2020) OFID.



>75
Enterokokken
Gram.-neg. Erreger



Erreger: abhängig von der Intervention

Kim et al., (2020) OFID.

mit Intervention: MRCNS

Organism	Patients With Previous Spinal Procedures (n = 128)	Patients Without Previous Spinal Procedures (n = 458)	PValue
<i>S. aureus</i>	48 (37.5)	207 (45.2)	.120
MSSA	33 (25.8)	124 (27.1)	.770
MRSA	15 (11.7)	83 (18.1)	.086
MSCNS	3 (2.3)	9 (2.0)	.730
MRCNS	11 (8.6)	8 (1.7)	.001
<i>Enterococcus</i> spp.	6 (4.7)	16 (3.5)	.598
<i>Streptococcus</i> spp.	28 (21.9)	90 (19.7)	.579
G(-) bacilli	23 (18.0)	107 (23.4)	.194
Anaerobes	3 (2.3)	8 (1.7)	.712
Polymicrobial	1 (0.8)	8 (1.7)	.692
Other	5 (3.9)	5 (1.1)	.045





Erreger: abhängig von der Grunderkrankung

Kim et al., (2020) OFID.

Diabetes: *S. aureus*

Leberzirrhose: gram-neg.

	Patients With DM (n = 180)	Patients Without DM (n = 406)	P	Patients With LC (n = 55)	Patients Without LC (n = 531)	P
<i>Staphylococcus aureus</i>	84 (46.7)	171 (42.1)	.306	14 (25.5)	241 (45.4)	.005
MSSA	49 (27.2)	108 (26.6)	.876	10 (18.2)	147 (27.7)	.130
MRSA	35 (19.4)	63 (15.5)	.240	4 (7.3)	94 (17.7)	.048
MSCNS	3 (1.7)	9 (2.2)	.764	1 (1.8)	11 (2.1)	1.000
MRCNS	5 (2.8)	14 (3.4)	.672	2 (3.6)	17 (3.2)	.696
<i>Enterococcus</i> species	10 (5.6)	12 (3.0)	.127	0 (0)	22 (4.1)	.252
<i>Streptococcus</i> species	34 (18.9)	84 (20.7)	.616	16 (29.1)	102 (19.2)	.082
Gram-negative bacilli	37 (20.6)	93 (22.9)	.527	18 (32.7)	112 (21.1)	.048
Anaerobes	3 (1.7)	8 (2.0)	1.000	3 (5.5)	8 (1.5)	.075
Polymicrobial	2 (1.1)	7 (1.7)	.729	1 (1.8)	8 (1.5)	.591
Other	2 (1.1)	8 (2.0)	.731	0 (0)	10 (1.9)	.610



Erreger: abhängig von der Grunderkrankung

Kim et al., (2020) OFID.

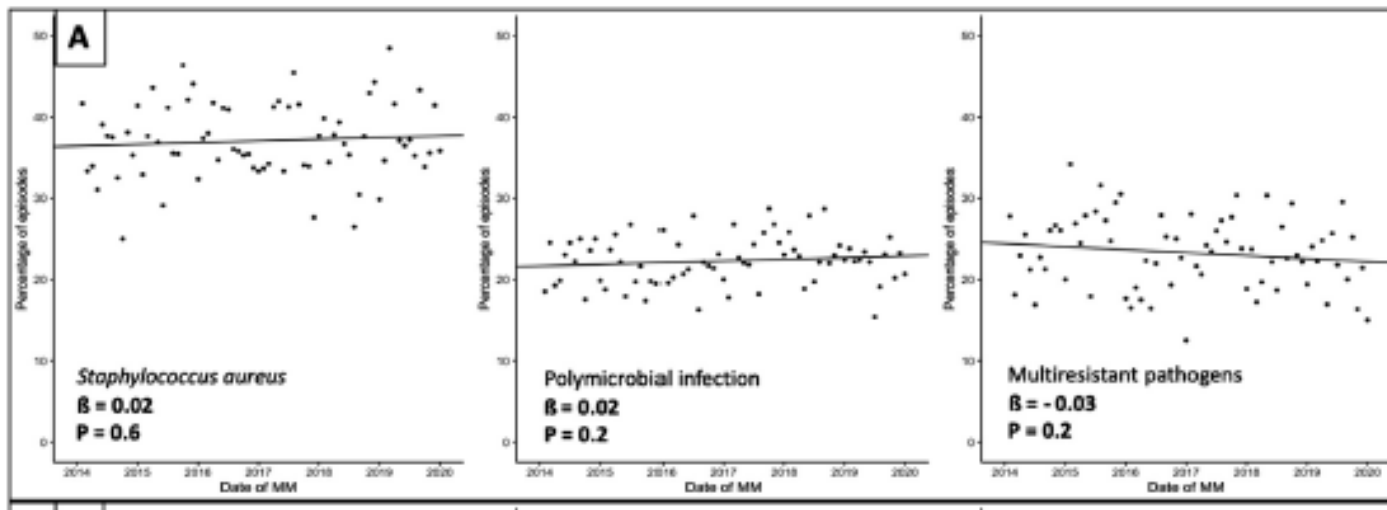
Niereninsuffizienz: MRSA

Tumor: gram-neg.

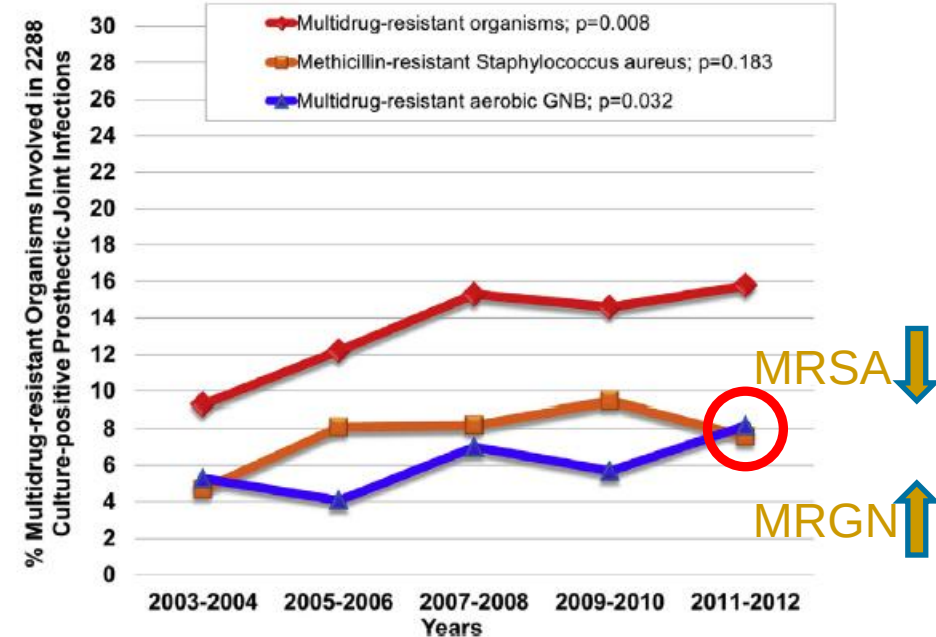
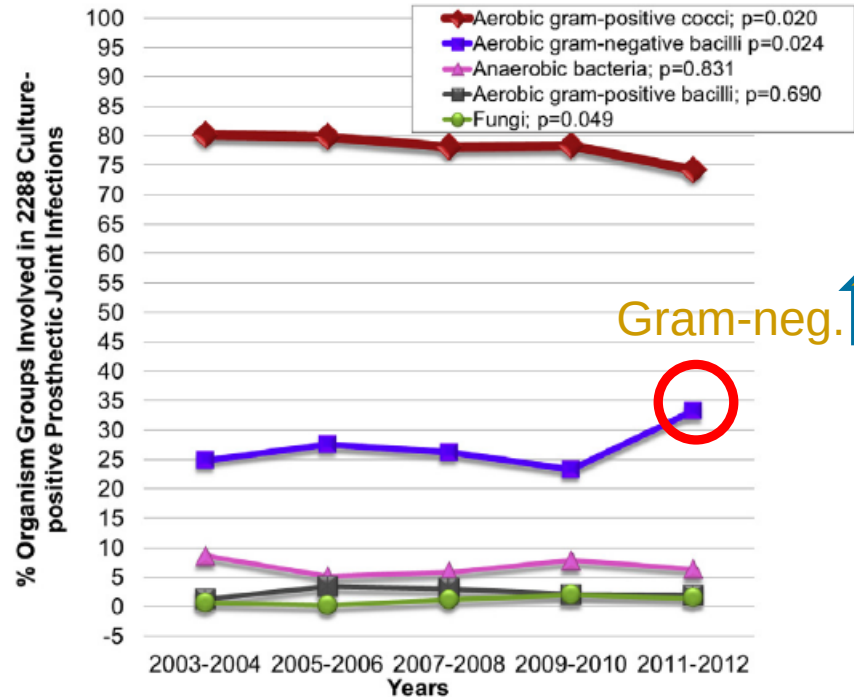
	Patients With CRD (n = 61)	Patients Without CRD (n = 525)	P	Patients With Solid Tumor (n = 84)	Patients Without Solid Tumor (n = 502)	P
<i>Staphylococcus aureus</i>	32 (52.5)	223 (42.5)	.14	27 (32.1)	228 (45.4)	.023
MSSA	11 (18.0)	146 (27.8)	.103	14 (16.7)	143 (28.5)	.024
MRSA	21 (34.4)	77 (14.7)	<.001	13 (15.5)	85 (16.9)	.741
MSCNS	1 (1.6)	11 (2.1)	>.99	3 (3.6)	9 (1.8)	.394
MRCNS	1 (1.6)	18 (3.4)	.709	1 (1.2)	18 (3.6)	.500
<i>Enterococcus</i> species	5 (8.2)	17 (3.2)	.068	1 (1.2)	21 (4.2)	.346
<i>Streptococcus</i> species	10 (16.4)	108 (20.6)	.441	23 (27.4)	95 (18.9)	.074
Gram-negative bacilli	10 (16.4)	120 (22.9)	.250	26 (31.0)	104 (20.7)	.037
Anaerobes	0 (0)	11 (2.1)	.615	0 (0)	11 (2.2)	.379
Polymicrobial	2 (3.3)	7 (1.3)	.239	2 (2.4)	7 (1.4)	.624
Other	0 (0)	10 (1.9)	.610	1 (1.2)	9 (1.8)	>.99

Verändert sich das Erregerspektrum über die Zeit?

Trends der Erreger **un**verändert!



Trends der Erreger verändert!



Fallvignette

- 72-jähriger Patient
- Vorerkrankungen:
 - Diabetes mellitus Typ II
 - Arterielle Hypertonie
 - Übergewicht – BMI 32 kg/m²
 - COPD
 - Vorhofflimmern (Markumar)
 - DDD-Schrittmacher
- Hüft-TEP-Einbau

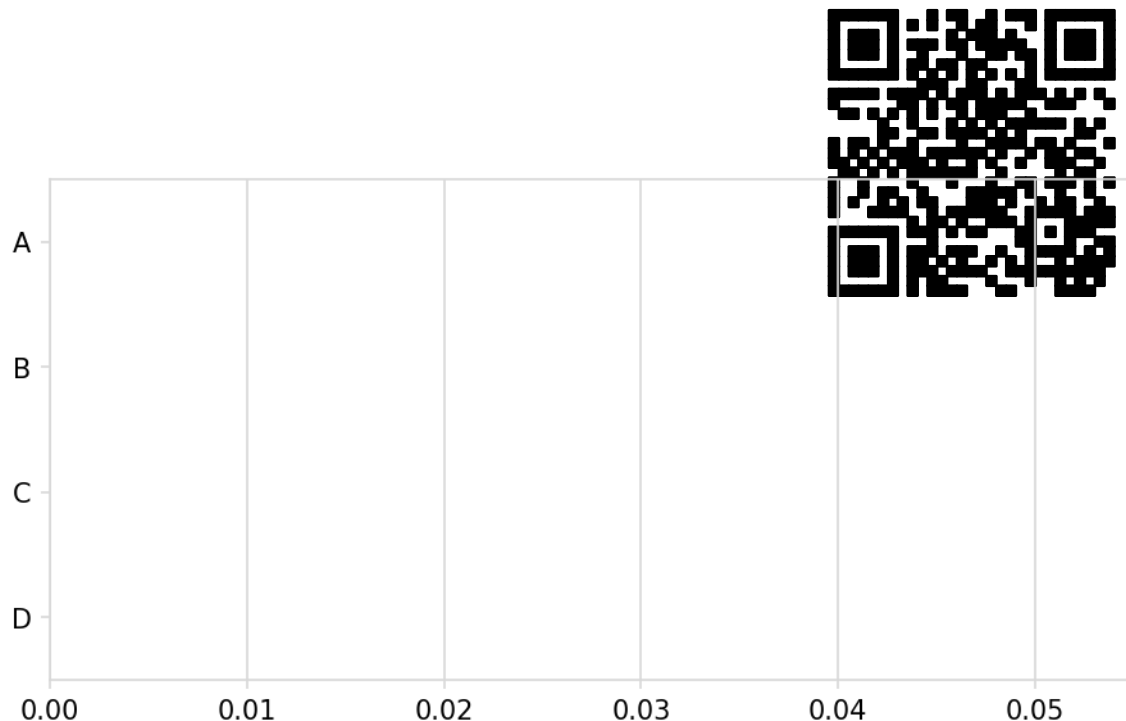
Wie hätten Sie den Patienten initial beim TEP Einbau aufgeklärt? Wie hoch schätzen Sie das Risiko eine PJI (90d) zu entwickeln?

A: 1,5 %

B: 3%

C: 5%

D: 8%



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Umfrage noch nicht

Modell zur Risikoabschätzung

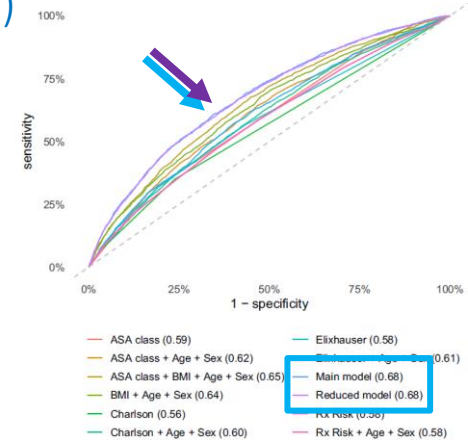
Bulow et al., (2022) Clin Epidemiol.

Receiver operating characteristics (ROC)

- Primäre THA Risiko für PJI (90 Tagen)
- 88.830 (Dänemark), 18.854 (Schweden)

- Faktoren:

Alter, Geschlecht, ASA, BMI, Vorerkrankungen (Arrhythmie, ZNS, Leber, Lunge, Elektrolytstörungen)





Kalkulator zur Risikoabschätzung

Bulow et al., (2022) Clin Epidemiol.
https://erikbulow.shinyapps.io/prediction_model/

RISKS WITHIN 90-DAYS AFTER TOTAL HIP REPLACEMENT

The average risk to die within 90 days after elective total hip arthroplasty is 0.3 % in Sweden (3 deaths for every 1,000 patients). There is a 2.5 % (1 in 40) risk for a prosthetic joint i

MANDATORY

Age

57 72 95

57 61 65 69 73 77 81 85 89 93 95

Sex

☒ ♂ ☐ ♀

ASA class

☒ I ☐ II ☐ III

☐ CNS disease

FOR SURVIVAL

Do you have any of the following:

☐ Cancer

☐ Kidney disease

☒ Diagnosed obesity

FOR INFECTION

Diagnosis

Primary osteoarthritis

BMI

class I obesity

Do you have any of the following:

☐ Fluid electrolyte disorders

☐ Liver disease

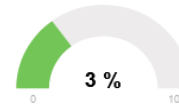
☐ Arrhythmia

☐ Lung and airways disease

Death Infection Information

Your probability

Your risk for a prosthetic joint infection within 90 days is close



Patients like you

It takes approximately 33 patients like you for one of them to h



Fallvignette

- 2 Jahre später
 - *S.aureus*-Sepsis
 - Schrittmacherexplantation
- 3 Monate später
 - Schmerzen rechte Hüfte
 - Fistel
 - Kein Fieber, keine Krankheitszeichen

Fallvignette

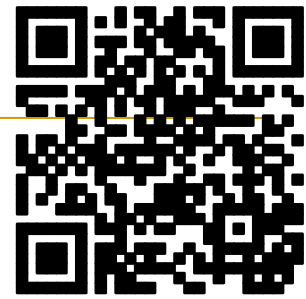
- Labor
 - CRP 60 mg/l, keine Leukozytose
- Probenentnahme:
 - Synovialflüssigkeit: *S.aureus*
 - SF: 2000 Leukozyten/ μ l
 - BK: negativ
- TEE
 - unauffällig

Nachweis von

1) Mäßig viel *Staphylococcus aureus*

<u>ANTIBIOGRAMM:</u>	<u>1</u>
Penicillin	R
Oxacillin	S
Ampicillin+Sulbactam	S
Piperacillin+Tazobactam	S
Cefazolin	S
Cefuroxim	S
Imipenem	S
Gentamicin	S
Ciprofloxacin	S
Moxifloxacin	S
Co-Trimoxazol	S
Tetracyclin	S
Erythromycin	I
Clindamycin	R
Vancomycin	S
Teicoplanin	S
Tigecyclin	S
Rifampicin	S
Fusidinsäure	S
Fosfomycin	S
Linezolid	S

Was ist allein ein sicheres („confirmed“) Diagnosekriterium für die PJI? (EBJIS)

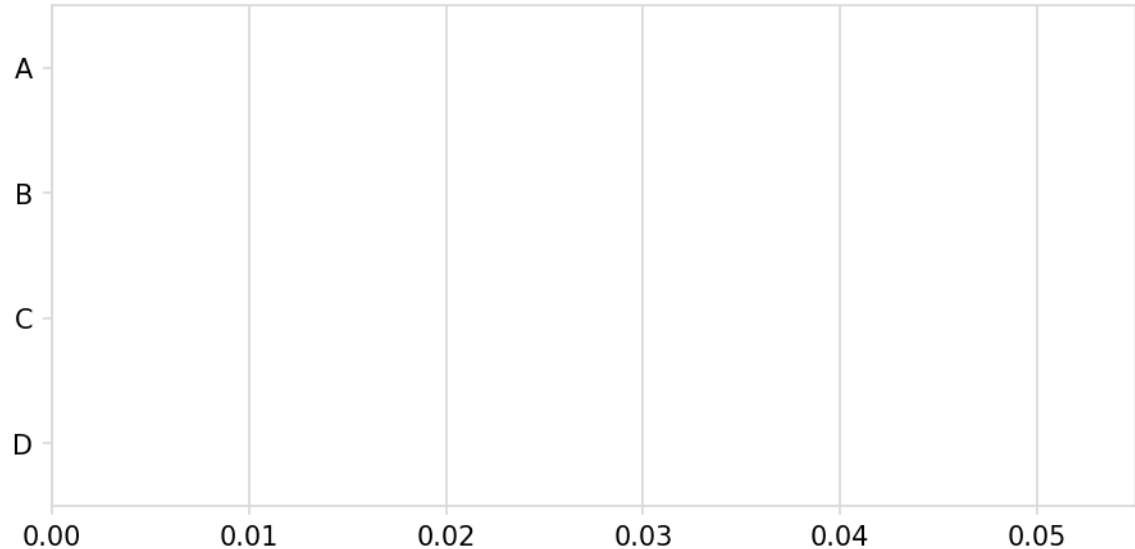


A: kürzlich zurückliegende
Blutstrominfektion

B: 2000 Leukoyten/ μ l in der
Synovialflüssigkeit

C: *S. aureus* in einer Probe

D: Sinustrakt mit
Kommunikation zum Gelenk



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Umfrage noch nicht

EBJIS Definition der PJI

Mc Nally et al.. (2021) Bone Joint J
al., (2023), NEJM

Table 3. Criteria for Diagnosis of Hip or Knee PJI.*

EBJIS “Confirmatory” Single Criteria⁵²

2018 Parvizi et al. “Major” Single Criteria⁵¹

Two positive cultures (includes synovial fluid, tissue, and sonicate-fluid cultures) for the same microorganism

Two positive cultures (includes synovial fluid, tissue, and sonicate-fluid cultures) for the same microorganism

Sinus tract with communication to joint or prosthesis

Sinus tract with communication to joint or prosthesis

Synovial fluid leukocyte count, >3000/ml†

Synovial fluid neutrophils, >80%†

Synovial fluid alpha-defensin positive‡

Sonicate-fluid culture, >50 CFU/ml for any organism (>200 CFU/ml if centrifuged)

Histopathological assessment (high-power field, 400x magnification) showing ≥5 neutrophils in ≥5 high-power fields (or visible microorganisms)§

2 positive Kulturen

Fistelgang zum Gelenk

Clinical and bla
Clinical feature
C-reactive prot
Synovial fluid
Leukocyte cou
PMN (%) ^c
Synovial fluid
Alpha-defensin
Microbiology ^f
Aspiration fluid
Intraoperative (fluid and tissu
Sonication ^h (Cl
Histology ^{c,j}
High-power fie (400x magnific

			Presence of visible microorganisms
Others			
Nuclear imaging	Negative three-phase isotope bone scan ^e	Positive WBC scintigraphy ^f	

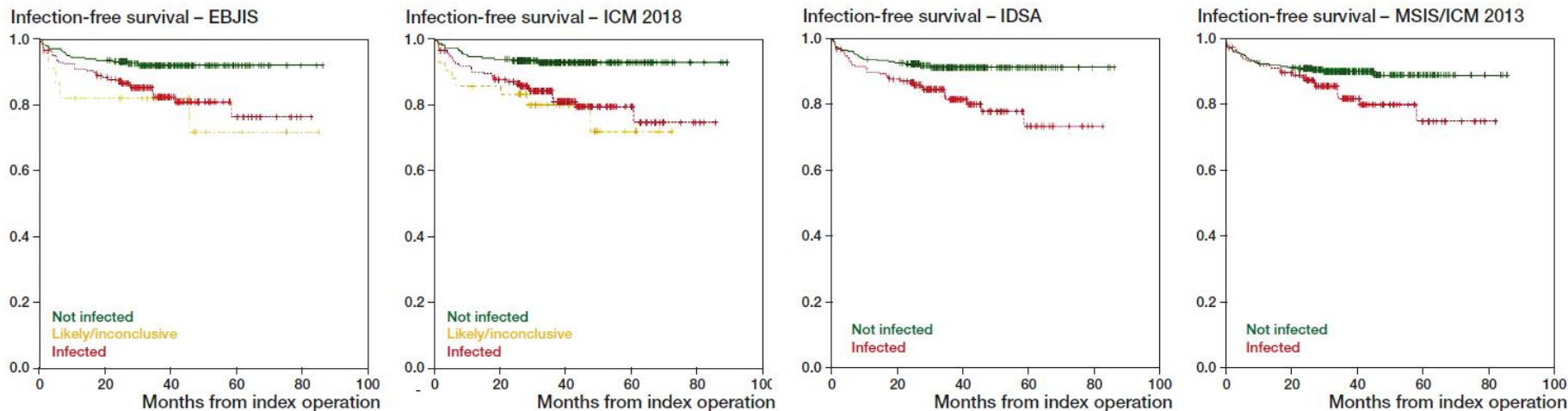


Figure 2. Kaplan–Meier infection-free survival curves for different classifications.

- EBJS – höchste Übereinstimmung:
präoperative verglichen mit definitiver Klassifizierung
(K = 0.9, CI 0.8-0.9)
- EBJS – besser zum Ausschluss
Infection unlikely:
NPV 90% (85-93)

Table 3. Final diagnosis classification according to different definitions. Values are count (%)

EBJIS	<i>Infection unlikely</i>	<i>Infection likely</i>	<i>Infection confirmed</i>
	255 (54)	22 (5)	195 (41)
ICM 2018	<i>Not infected</i>	<i>Inconclusive</i>	<i>Infected</i>
	258 (55)	42 (9)	172 (36)
IDSA	<i>Not infected</i>	–	<i>Infected</i>
	284 (60)	–	188 (40)
MSIS 2013	<i>Not infected</i>	–	<i>Infected</i>
	327 (69)	–	145 (31)

Internationale einheitliche Diagnosekriterien
in Planung!

Warum würden Sie die Prothese a.e. entfernen?

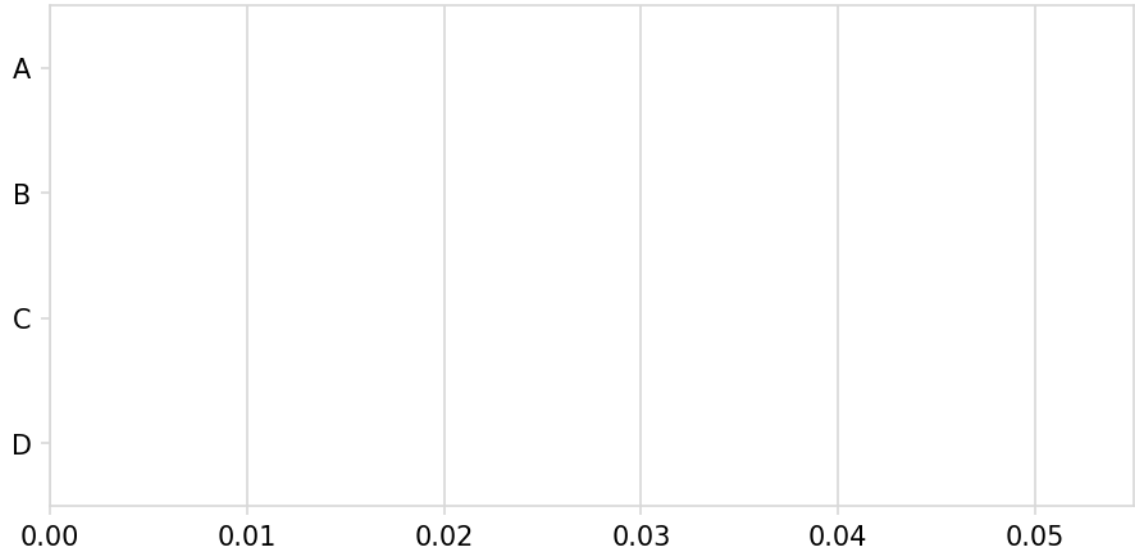


A: Nachweis von *S. aureus*

B: chronische Infektion

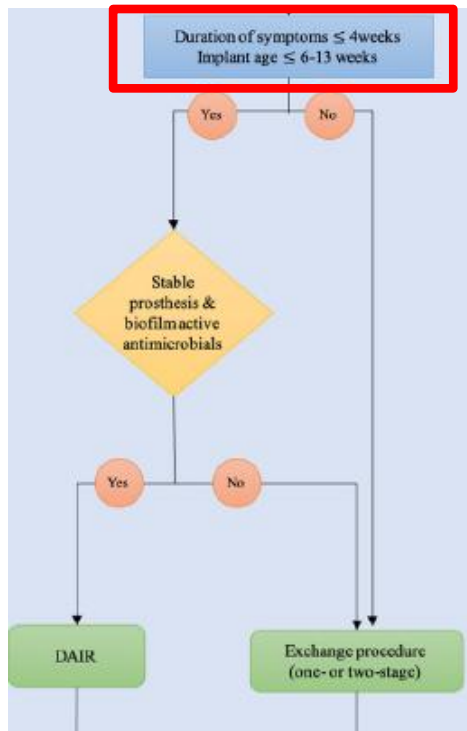
C: späte Infektion

D: Z.n Blutstrominfektion

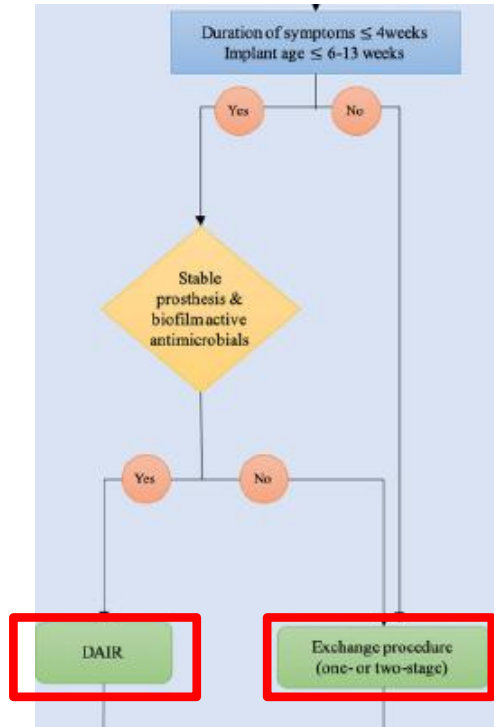


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Umfrage noch nicht



- **Erhalt: akut, früh, stabil, Weichteile intakt, biofilm-aktives Antibiotikum**
- Akut: 3-4 Wochen Symptomdauer
- **Chronisch: > 3-4 Wochen Symptomdauer**
- **Standard – Explanation!**
- Früh: < 12 Wochen seit Implantation
- Spät: > 12 Wochen seit Implantation
- Unklar?

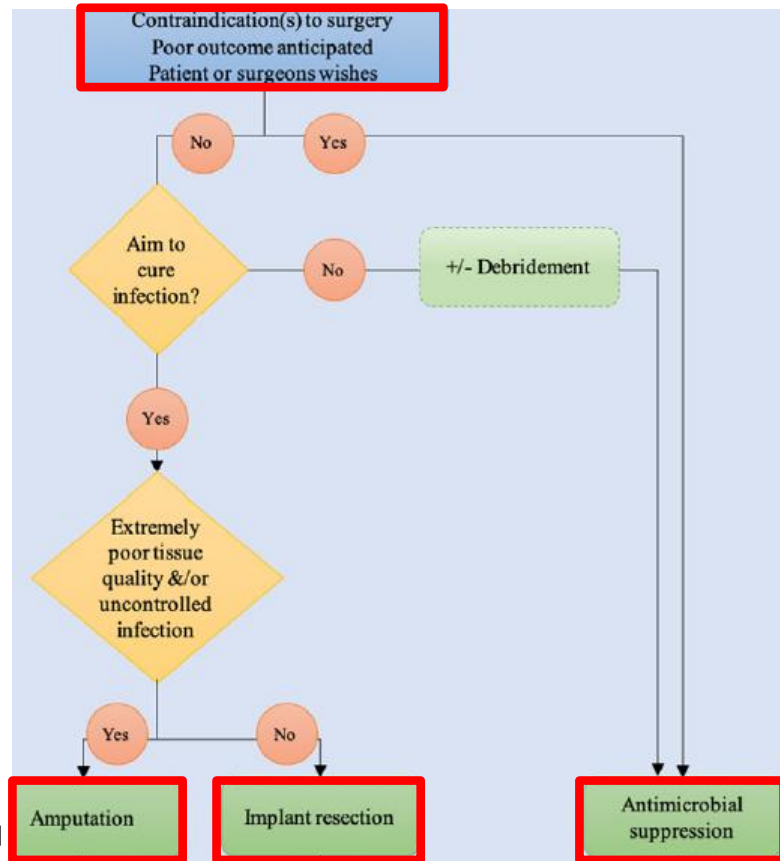


- DAIR (*Debridement, Antibiotics, Implant Retention*)
- 1-zeitiger Wechsel
- 2-zeitiger Wechsel –mit kurzem oder langem Intervall

Chirurgisches Vorgehen – kein Heilung oder Erhalt der Funktionalität



Peel et al., (2020) CMI



- Amputation
- Resektion ohne erneuten Einbau
- DAIR –mit Suppressionstherapie

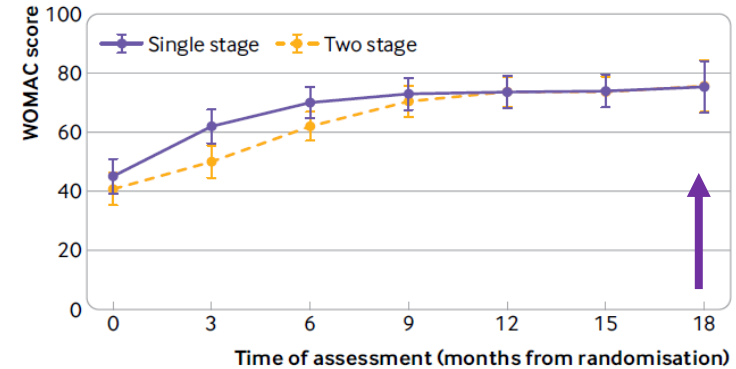
Ein- oder zweizeitiger Wechsel? - Lebensqualität

- ein- (n=65) versus zweizeitig (n=75)
- Primärer Endpunkt: Schmerz, Steifigkeit, Funktionseinschränkungen nach 18 Monaten (WOMAC Score)
- Sekundäre Ziele: chirurg. Komplikationen, Gelenkinfektionen, ökonomische Evaluation (QUALYs)

Ein- oder zweizeitiger Wechsel? - Lebensqualität

Blom et al., (2022) BMJ

- Primärer Endpunkt:
 - WOMAC Score: kein Unterschied
- Sekundäre Endpunkte: 1-versus 2- zeitig
 - Intraoperative Komplikationen geringer (8% versus 27%)
 - Infektionen gleich (14% versus 11%)
 - Nettonutzen £11.167 (£ 20.000 pro QUALY)

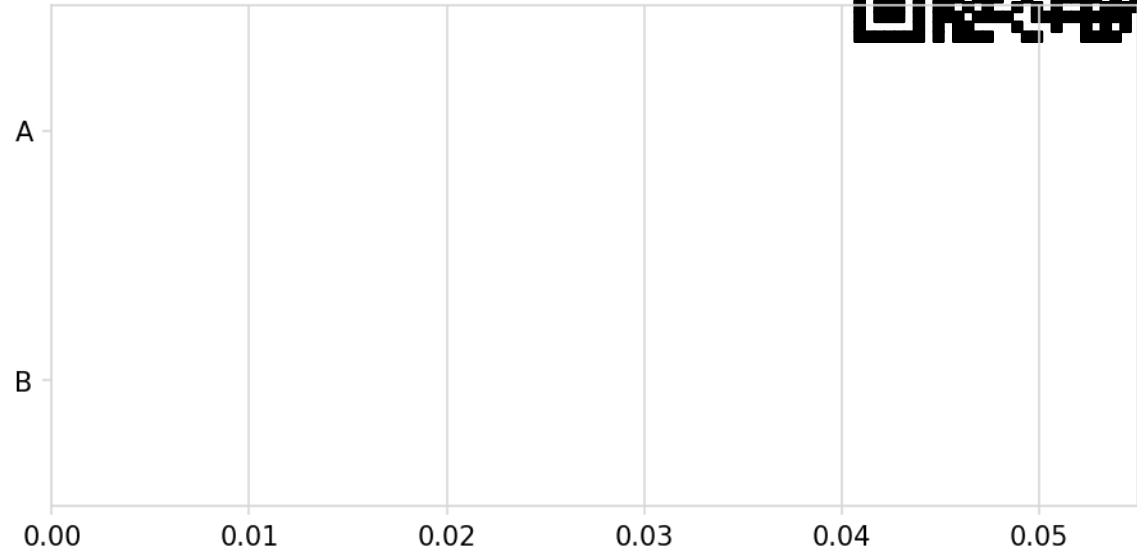


Sie planen den Ausbau und möchten Proben gewinnen – wann geben Sie die präoperative Prophylaxe?



A: präoperativ

B: intraoperativ



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Umfrage noch nicht

ANTIBIOTICS WITH LOW PREOPERATIVE SUSPICION OF PJI OR ESTABLISHED PJI WITH A KNOWN PATHOGEN

Update of 2009 CPG Recommendation

Strong evidence supports that preoperative prophylactic antibiotics be given prior to revision surgery in patients at low preoperative suspicion for periprosthetic infection and those with an established diagnosis of periprosthetic joint infection of known pathogen who are undergoing reoperation.

Strength of Recommendation: Strong 

Description: Evidence from two or more “High” quality studies with consistent findings for recommending for or against the intervention.

Knie-Revisionseingriffe: mit oder ohne perioperative Prophylaxe



Wouthuyzen-Bakker, Journal of Arthroplasty, 2017
Patel et al., 2023 NEJM

- Retrospektiv
- Revisionschirurgie Knie-TEP: aseptische Lockerung/chronische Schmerzen
 - mit (n= 141)/ohne (n= 284) perioperative Prophylaxe
- Ziel:
 - Rate positiver Kulturen
 - Inzidenz von PJI innerhalb von 2 Jahren

Knire-Revisionseingriffe: ohne perioperative Prophylaxe – Risiko von Infektionen erhöht



Wouthuyzen-Bakker, Journal of Arthroplasty, 2017
Patel et al., 2023 NEJM

- Ergebnisse
 - Mind. 1 pos. Kultur
 - ohne (26%)
 - mit (27%); $p=0,07$
 - PJI (MISC):
 - ohne (6,7%)
 - mit (7,0%); $p= 0,79$
 - Mikroorganismen vergleichbar
 - Frühe PJI –innerhalb 3 Monate
 - ohne (6,4%)
 - mit (1,6%) ; $p=0,01$

The culture yield is likely to be higher when antibiotics are withheld for at least 2 weeks before culture.¹⁸ However, prophylactic preoperative antibiotic therapy does not reduce the culture yield and should therefore be administered.

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Zusammenfassung

- Grobe Risikoeinschätzung für PJI
- Prothesenerhalt –
 - v.a. akut (3 Wochen), früh (3 Monate), stabil, Weichgewebe intakt, bei Staphylokokken Rifa S
- Prothesenwechsel-
 - Einzeitig möglicherweise von Vorteil
- Unterschiedliche internationale Diagnosekriterien:
 - EBJIS – in der Validierung am besten
 - Einheitliche international Kriterien in Planung
- Perioperative Prophylaxe vor Probengewinnung?!
 - bei allen?

Fallvignette: Prothesenerhalt nicht möglich

- Explantation
- Start mit Cefazolin 3x2g i.v.
- Wie geht es weiter?

Fallvignette: Wie lange würden Sie therapieren?

- A) 6 Wochen
- B) 12 Wochen
- C) 24 Wochen

Fallvignette: Wann würden Sie oralisieren?

- A) Abhg. vom Ansprechen
- B) Nach 1 Woche
- C) Nach 2 Wochen

Häufige Fragen

- Wie lange therapieren?
- Wann oralisieren?
- Womit oralisieren?



- Eindeutige Empfehlung nur bei
- Eindeutiger Evidenz
- Diskussion verschiedener Optionen, wenn Evidenz fehlt
- ↔ Grade: strong/weak möglich

Häufige Fragen

- Wie lange therapieren?
- Wann oralisieren?
- Womit oralisieren?

Therapielänge – Spondylodiszitis - Landmarkstudie



Bernard et al., (2015) Lancet
Berbari et al., (2015), CID

Antibiotic treatment for 6 weeks versus 12 weeks in patients with pyogenic vertebral osteomyelitis: an open-label, non-inferiority, randomised, controlled trial

Louis Bernard, Aurélien Dinh, Idir Ghout, David Simo, Valerie Zeller, Bertrand Issartel, Vincent Le Moing, Nadia Belmatoug, Philippe Lesprit, Jean-Pierre Bru, Audrey Therby, Damien Bouhour, Eric Dénes, Alexa Debaré, Catherine Chirouze, Karine Fèvre, Michel Dupon, Philippe Aegerter, Denis Mulleman, on behalf of the Duration of Treatment for Spondylodiscitis (DTS) study group*

	6-week regimen	12-week regimen	Difference in proportion of patients*	95% CI
Intention-to-treat analysis, n	176	175		
Cured	160 (90.9%)	159 (90.9%)	+0.1	-6.2 to 6.3
Cured and alive†	156 (88.6%)	150 (85.7%)	+2.9	-4.2 to 10.1
Cured without further antibiotic treatment‡	142 (80.7%)	141 (80.6%)	+0.1	-8.3 to 8.5
Per-protocol analysis, n	146	137		
Cured	137 (93.8%)	132 (96.4%)	-2.5	-8.2 to 2.9
Cured and alive†	133 (91.1%)	126 (92.0%)	-0.9	-7.7 to 6.0
Cured without further antibiotic treatment‡	NA	NA	NA	NA

- 6 versus 12 Wochen kein Unterschied
- Bei nativer Spondylodiszitis: 6 Wochen Standard

Therapielänge – periprotetische Infektion - Landmarkstudie



Bernard et al., (2021) NEJM

THE NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Antibiotic Therapy for 6 or 12 Weeks for Prosthetic Joint Infection

L. Bernard, C. Anvieux, B. Brunschweiler, S. Touchais, S. Ansart, J.-P. Bru, E. Oziol, C. Boeri, G. Gras, J. Druon, P. Rosset, E. Senneville, H. Bentayeb, D. Bouhour, G. Le Moal, J. Michon, H. Aumaitre, E. Forestier, J.-M. Laffosse, T. Begué, C. Chirouze, F.-A. Dauchy, E. Devaud, B. Martha, D. Burgot, D. B. E. Stindel, A. Dinh, P. Berner, B. Giraudeau, B. Issartel, and A. C.

- 6 Wochen- höhere Rate
- Non-inferiority: nicht ge
- Standard 12 Wochen !

Analysis	6-Wk Therapy no. of patients with event/total no. (%)	12-Wk Therapy no. of patients with event/total no. (%)	Risk Difference Percentage points (95% CI)	Adjusted Risk Difference* Percentage points (95% CI)
Modified intention-to-treat				
Main analysis in which missing outcomes for patients who were lost to follow-up were considered to be persistent infections and data from patients who died removed†	35/193 (18.1)	18/191 (9.4)	8.7 (1.8–15.6)	9.0 (2.3–15.7)

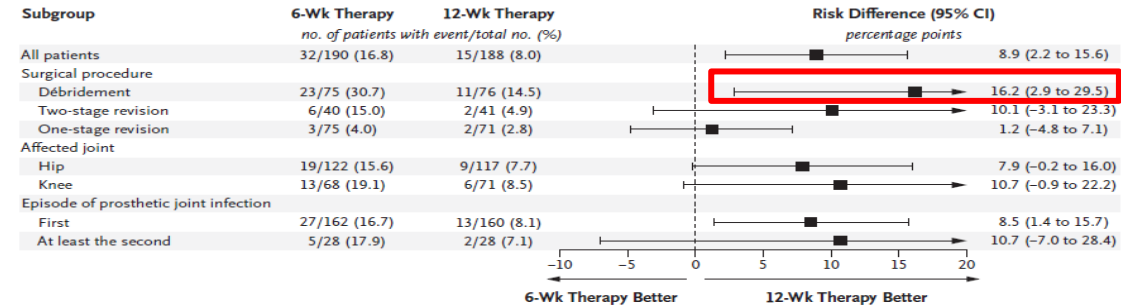


Figure 2. Exploratory Subgroup Analyses of Persistent Infection within 2 Years after the Completion of Antibiotic Therapy (Primary Outcome).

Therapielänge – diabetische Fußinfektion- randomisierte Studie



Tone et al., (2015) Diabetes care



Six-Week Versus Twelve-Week Antibiotic Therapy for Nonsurgically Treated Diabetic Foot Osteomyelitis: A Multicenter Open-Label Controlled Randomized Study

Diabetes Care 2015;38:302–307 | DOI: 10.2337/dc14-1514

Table 4—Clinical outcome of 40 diabetic patients with osteomyelitis of the foot treated nonsurgically according to the duration of antibiotic therapy

Patient outcome	6 weeks <i>n</i> = 20	12 weeks <i>n</i> = 20	<i>P</i>
Overall remission	12 (60)	14 (70)	0.50
Complete healing ^a	18 (90)	16 (80)	0.38
Time to complete healing (weeks ± SD)	13.1 ± 12.2	16.8 ± 17.4	0.44
Overall failure	8 (40)	6 (30)	0.50
Noncomplete healing	2 (10)	4 (20)	0.37
Relapsing osteomyelitis	2 (15)	3 (15)	1
Worsening radiological bone abnormalities	6 (30)	4 (20)	0.46
Bone resection	2 (10)	2 (10)	1
Spread of osteomyelitis to contiguous sites	4 (20)	2 (10)	0.37
Major amputation	2 (10)	2 (10)	1

Data are number of patients in whom the event was recorded (%). The total number of each column may exceed the total (overall failure), since more than one event may be recorded in a given patient. ^aComplete healing of wound sustained for at least 4 consecutive weeks.

- 6 Wochen vergleichbar mit 12 Wochen

Therapielänge – diabetische Fußinfektion- laufende Studien



Waibel et al., (2020) Trials

- Nicht-operiert
 - 3 versus 6 Wochen
- Debridement /Amputation– Weichgewebe:
 - 10 versus 20 Tage/1 versus 4 Tage
- Debridement/Amputation – Knochen:
 - 3 versus 6 Wochen/1 versus 3 Wochen

Therapielänge – Knocheninfektionen (Wiki)

Spellberg et al., (2022) JAMA Network Open

Table 4. Summary of Antibiotic Durations for Osteomyelitis

Condition	Clear recommendation	Clinical review
Osteomyelitis without retained implant (including DFO)	Maximum 6 wks	3-4 wks may be adequate with debridement; confirmatory studies desired
Osteomyelitis with total resection of infected bone	None	No antibiotics is a reasonable option; not recommended for use exceeding 5 d
PJI with DAIR	None	All participating experts preferred 12 wks; a confirmatory, second study is needed to enable a clear recommendation
PJI with exchange	None	12 wks favored by some experts - Other experts believed equipoise remains for 6 vs 12 wks 6 wks may be reasonable for non-<i>S aureus</i> pathogens, particularly for 1-stage exchanges 6 wks may be reasonable for 2-stage exchange, although there is controversy about the need for further antibiotics after the second stage (reimplantation)

Häufige Fragen

- Wie lange therapieren?
- Wann oralisieren?
- Womit oralisieren?

Oralisierung– Knocheninfektionen - randomisierte Studie



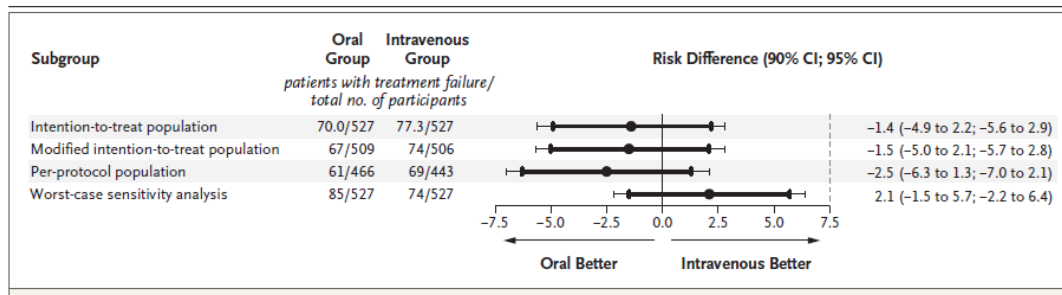
Li et al., (2019) NEJM

THE NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Oral versus Intravenous Antibiotics for Bone and Joint Infection

H.-K. Li, I. Rombach, R. Zambellas, A.S. Walker, M.A. McNally, B.L. Atkins, B.A. Lipsky, H.C. Hughes, D. Bose, M. Kümin, C. Scarborough, P.C. Matthews, A.J. Brent, J. Lomas, R. Gundle, M. Rogers, A. Taylor, B. Angus, I. Byren, A.R. Berendt, S. Warren, F.E. Fitzgerald, D.J.F. Mack, S. Hopkins, J. Folb, H.E. Reynolds, E. Moore, J. Marshall, N. Jenkins, C.E. Moran, A.F. Woodhouse, S. Stafford, R.A. Seaton, C. Vallance, C.J. Hemsley, K. Bisnauthsing, J.A.T. Sandoe, I. Aggarwal, S.C. Ellis, D.J. Bunn, R.K. Sutherland, G. Barlow, C. Cooper, C. Geue, N. McMeekin, A.H. Briggs, P. Sendi, E. Khatamzas, T. Wangrangsimekul, T.H.N. Wong, L.K. Barrett, A. Alvand, C.F. Old, J. Bostock, J. Paul, G. Cooke, G.E. Thwaites, P. Bejon, and M. Scarborough, for the OVIVA Trial Collaborators*



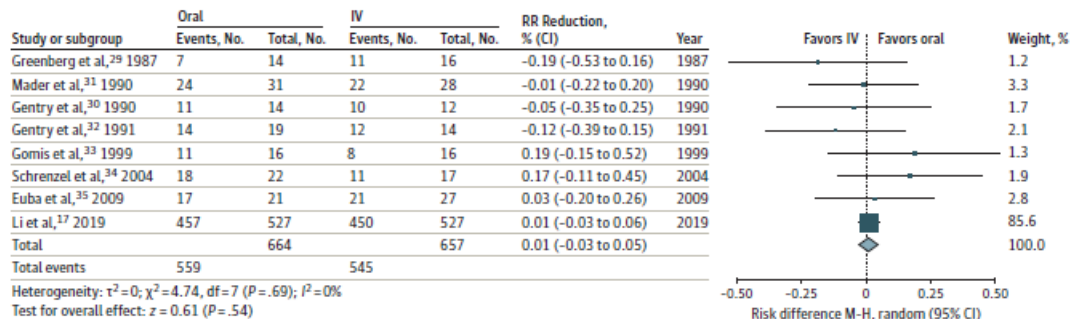
- Definitives Therapieversagen nach einem Jahr:
- 14.6% i.v. - 13.2% oral – Delta: 1.4% - (95% CI -5.6-2.9)
- Frühe Oralisierung nicht unterlegen (max. d7)

Oralisierung– Knocheninfektionen (Wiki)



Spellberg et al., (2022) JAMA Network Open
Sendi et al., (2023), CMI

Figure. Random-Effects Meta-analysis Forest Plot of Randomized Clinical Trials Comparing Long-term Clinical Success Rates of Oral vs Intravenous (IV) Antibiotic Therapy for Osteomyelitis in Adults



- Klinische Entscheidung
- Am Bett
- Multidisziplinär

- klinisch stabil, Fokus adäquat saniert, gute orale Resorption, Regime mit Evidenz, Adhärenz
- Keine Mindestdauer



Positiv- Empfehlungen

Bei fehlender klinischer Kontraindikation sollen orale statt intravenöse Antibiotika mit guter oraler Bioverfügbarkeit appliziert werden.



Don't prescribe intravenous (IV) antibiotics for patients who can safely be treated with an oral option, given that IV antibiotics have a higher carbon footprint.

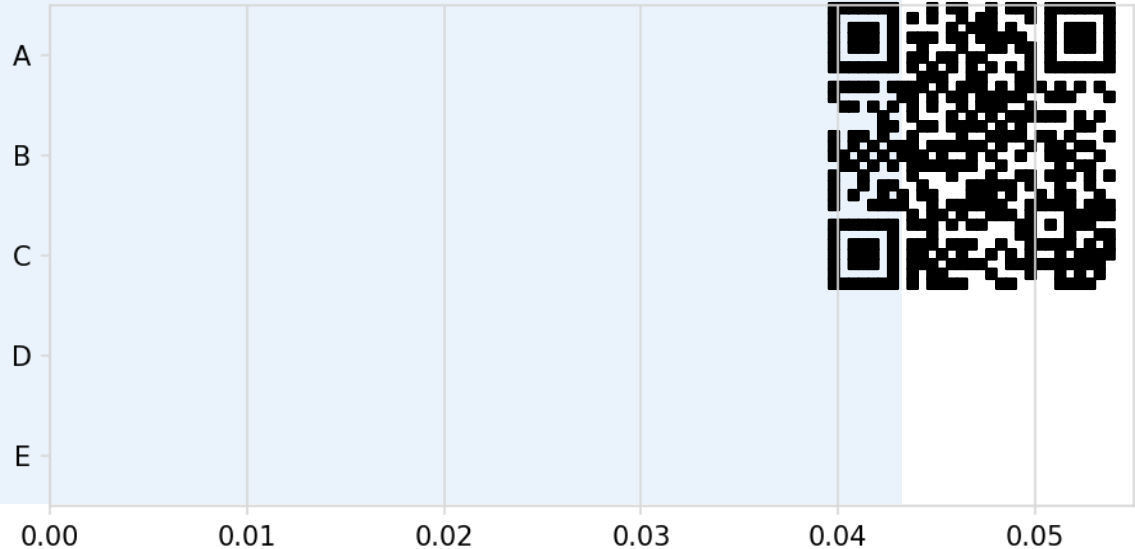
1 Woche Ciprofloxacin (oral: CO₂e: 1,4 kg)

Wieviel höher ist das CO₂-Äquivalent für i.v. Ciprofloxacin im Vergleich zur p.o. Gabe?



<https://doi.org/10.1136/bmjqs-2023-015928>
Born et al., BMJ, 2023,12 Internal medicine
<https://choosingwiselycanada.org/climate/>

- A) + 20 %
- B) + 50 %
- C) 5-fach
- D) 10-fach
- E) 100-fach



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Umfrage noch nicht



Don't prescribe intravenous (IV) antibiotics for patients who can safely be treated with an oral option, given that IV antibiotics have a higher carbon footprint.

1 Woche Ciprofloxacin oral (1,4 kg CO₂e) – 6,8 km Auto

1 Woche Ciprofloxacin intravenös (100,1 kg CO₂e) – 485,9 km Auto

Häufige Fragen

- Wie lange therapieren?
- Wann oralisieren?
- Womit oralisieren?

Praxistipps für die Oralisierung

Mit freundl. Genehmigung; Dr. Weber, Prof. Dränert
Copyright ABS Team LMU

Bei fehlender klinischer Kontraindikation sollen **orale statt intravenöse Antibiotika** mit guter oraler Bioverfügbarkeit appliziert werden.

Oralisierung immer möglich:

- Klinische Besserung, Patient hämodynamisch stabil
- Resorption und orale Aufnahme sind nicht beeinträchtigt
- Geeignetes Antibiotikum mit ausreichend hoher oraler Bioverfügbarkeit verfügbar

- Ciprofloxacin, Moxifloxacin
- Cotrimoxazol
- Linezolid
- Metronidazol
- Clindamycin
- Rifampicin
- Doxycyclin

← Direkte Umstellung auf orale Darreichungsform möglich

- Ampicillin ± Sulbactam (iv) → Amoxicillin ± Clavulansäure (po)
- Penicillin G bzw. V
- Clarithromycin, Azithromycin

← Umstellung auf orale Darreichungsform abhängig von Indikation

- Cefuroxim
- Flucloxacillin
- Fosfomycin
- Vancomycin
- Colistin
- Gentamicin, Amikacin, Tobramycin

← Direkte Umstellung auf orale Darreichungsform NICHT möglich

Table 3. Summary of Oral Antibiotic Doses Used in Published Studies for Osteomyelitis

Drug	Dose	Comments
Ciprofloxacin	500-750 mg twice daily	Higher dose for pseudomonas
Levofloxacin	750 mg once daily	L-enantiomer of ofloxacin, the latter of which was widely studied for osteomyelitis
TMP-SMX	7.5-10 trimethoprim mg/kg/d divided twice or thrice daily (eg, 2 DS tablets twice daily for a 70 kg adult)	Most studies used 7.5-10 mg/kg/d, 2 studies used 4-6 mg/kg/d, with lower cure rates in 1 of them
Clindamycin	600 mg 3 times/d; 900 mg 3 times/d or 600 mg 4 times/d for larger patients	450 mg 4 times/d may be used but was not favored in published studies
Linezolid	600 mg twice daily	Standard dosing, monitor for reversible hematotoxicity after 2 weeks, and irreversible neurotoxicity after 4 wks
Amoxicillin/clavulanate	500 mg 3 times/d or 875 mg twice daily	Specifically for DFO
Rifampin	600 mg once daily	Doses studied include 600 once daily, 900 mg once daily or 600 mg twice daily, unclear if efficacy or toxicity differs; 300 mg doses may be less desirable due to lower AUC levels and less convenience for patients
Fosfomycin ^a	4-16 g per day	Various doses studied with formulations available outside the US, not studied with the sachet powder formulation in the US

CAVE !

- Chinolone



- Rifampicin

Currently displaying a list of **508 drugs** known to interact with rifampin.

- 206 major drug interactions
- 243 moderate drug interactions
- 59 minor drug interactions



Negativ- Empfehlungen

Rifampicin interagiert mit vielen Medikamenten. Es soll vor allem nicht gleichzeitig mit NOAKs verabreicht werden.



Spellberg et al. (2025)
American Journal of Medicine
Spellberg et al., (2022)
JAMA Network Open
Montori VM et al.(2023)
BMJ Evidence-Based
Medicine

2025 Update to WikiGuidelines for the management of pyogenic osteomyelitis in adults

Executive Summary of Clear Recommendations or Clinical Review Conclusions for Osteomyelitis.....	7
1. How should the diagnosis of osteomyelitis be established?	7
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3. When should empiric therapy be administered in the treatment of osteomyelitis?	8
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6. What is the role and optimal utilization of serial biomarkers and/or imaging studies for assessing treatment response in osteomyelitis?	10
7. What is the appropriate duration of therapy for typical cases of osteomyelitis?	10
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Shared decision-making as a method of care



Spellberg et al. (2025)
American Journal of Medicine
Spellberg et al., (2022)
JAMA Network Open

Rifampicin –wie lange?

If rifampin does reduce treatment failure rate for infected prosthetic joints, a recent observational study suggested that outcomes were similar in patients treated with rifampin 300 mg twice daily for twelve weeks versus patients treated with rifampin 600 mg twice daily for only 5 days.⁴⁹⁶ These results suggest that if rifampin is effective, shorter durations with higher doses might result in similar outcomes to longer durations with lower doses. Nevertheless, these results are hypothesis-generating, and a randomized controlled trial is desired to elucidate rifampin dosing and duration.

Optionen für MDR-GN Knocheninfektionen – individuelles Vorgehen



Tsilika et al., (2025) Pathog



Review

Antimicrobial Treatment Options for Gram-Negative Pathogens in Bone Infections

Maria Tsilika ¹, Fotinie Ntziora ² and Efthymia Giannakopoulou ³

Optionen:

Carpeneme

Meropenem-Vaborbactam

Imipenem/Cilastatin-Relebactam

Ceftolozan/Tazobactam

Ceftazidim/Avibactam

Cefiderocol

Fosfomycin

Colistin

Tigecyclin

Table 1. Highlights of antibiotics for multidrug-resistant (MDR) Gram-negative bacteria (GNB).

Antibiotic	Mechanism of Action	Spectrum of Activity of Antibiotics for MDR GNB	Limitations
Carbapenems	Inhibits penicillin-binding proteins (PBPs) in the cell wall synthesis of GNB	Broad spectrum, including MBL-, DTR <i>P. aeruginosa</i> , DTR <i>A. baumannii</i>	No activity against carbapenemases, DTR <i>A. baumannii</i> , DTR <i>P. aeruginosa</i>
			Ertapenem is inactive against <i>P. aeruginosa</i>
			No activity against MBL- or OXA-type carbapenemases, DTR <i>P. aeruginosa</i> , DTR <i>A. baumannii</i>
			No activity against MBL
Ceftolozan/Tazobactam	Inhibits penicillin-binding proteins (PBPs) in the cell wall synthesis of GNB	Broad spectrum, including MBL-, DTR <i>P. aeruginosa</i> , DTR <i>A. baumannii</i>	No activity against carbapenemases, DTR <i>A. baumannii</i> , AmpC β -lactamases
			No activity against MBL, High resistance rates in <i>A. baumannii</i>
Ceftazidim/Avibactam	Inhibits penicillin-binding proteins (PBPs) in the cell wall synthesis of GNB	Broad spectrum, including MBL-, DTR <i>P. aeruginosa</i> , DTR <i>A. baumannii</i>	No activity against MBL, High resistance rates in <i>A. baumannii</i>
			Its use as monotherapy is generally not recommended
Cefiderocol	Inhibits penicillin-binding proteins (PBPs) in the cell wall synthesis of GNB	Broad spectrum, including MBL-, DTR <i>P. aeruginosa</i> , DTR <i>A. baumannii</i>	It is recommended to be used in combination with one or more additional agents to which the pathogen displays a susceptible MIC
			It is not effective against <i>P. aeruginosa</i>
Tigecycline	Inhibits protein synthesis by binding to the bacterial 30S ribosomal subunit	ESBL-, CRE (all classes of carbapenemases, including MBL), DTR <i>A. baumannii</i>	It is not effective against <i>P. aeruginosa</i>



Zusammenfassung II

- Gram-neg. Knocheninfektionen steigend
- Weltweit – Multiresistenz steigend
- Rationale Therapie – Start nach Diagnostik
- Wie lange therapieren?
 - Osteomyelitis, VO – 6 Wochen
 - PJI – 12 Wochen (DAIR) (z.T. 6 Wochen erwägen)
- Wann oralisieren?
 - i.d. Regel keine Mindest-i.v. Dauer
 - Multidisziplinär entscheiden – bei gutem Ansprechen

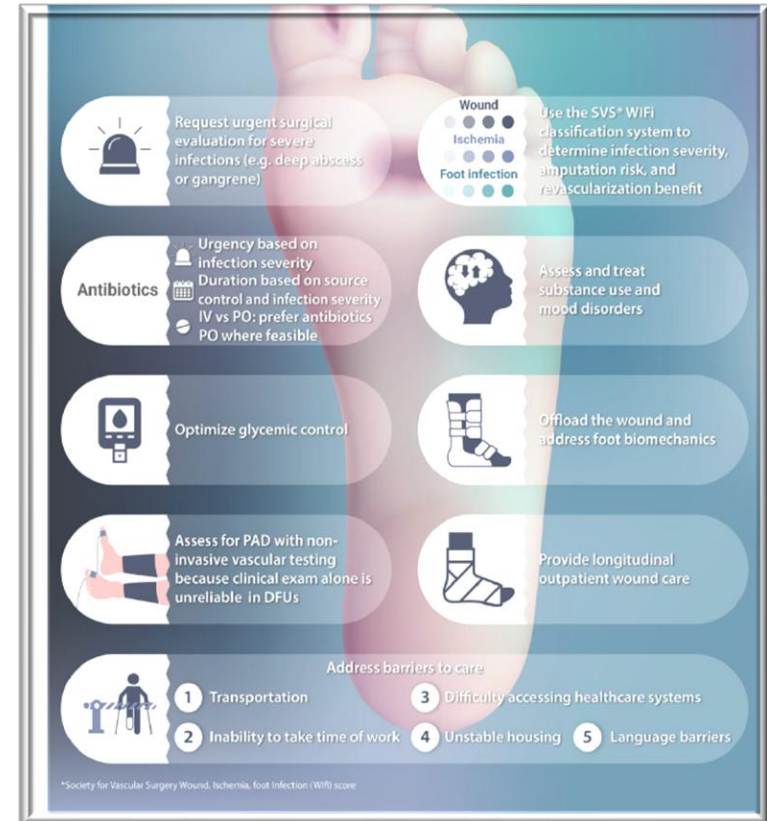
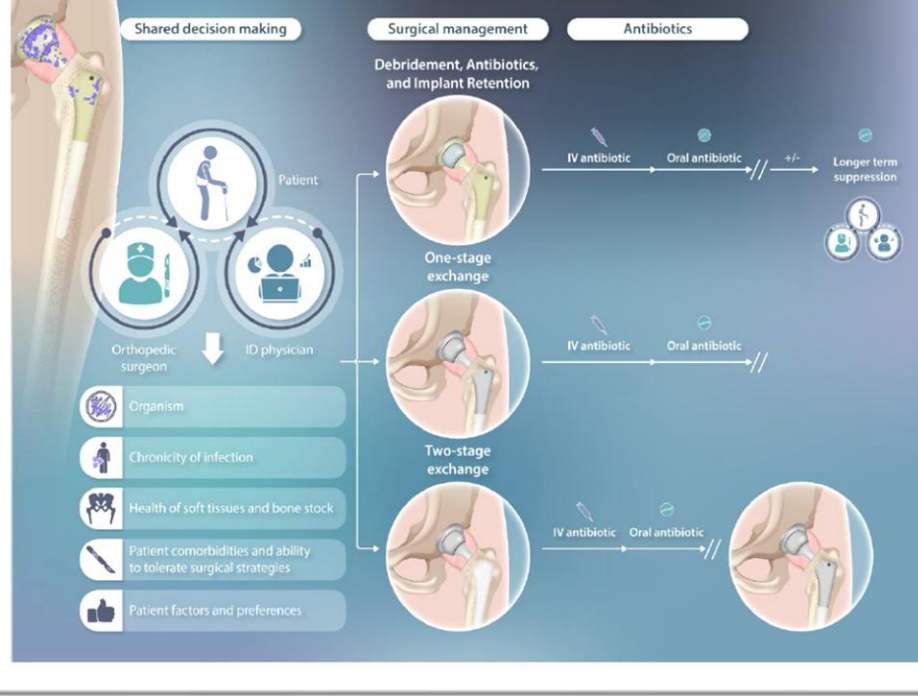
Zusammenfassung

- Womit oralisieren?
 - Bioverfügbarkeit und oral mögliche Dosis
 - Rote Handbrief (Chinolone)
 - Rifampicin – viele Fragen
- Antibiotikatherapie und Klima
 - Oral deutlich geringeres CO₂-Äquivalent
- Therapieoptionen für MRE
 - Einige neue Substanzen auf dem Markt
 - Individuelles Vorgehen

Knocheninfektionen – interdisziplinäres Management

Patel et al., (2023), NEJM
Cortes-Penfiled, (2023), CID

Periprosthetic Joint Infection



The End?



akademie für
infektionsmedizin e.V.

Startseite > Kurse & Veranstaltungen > Infektiologie Kursus: Knochen- und Gelenkinfektionen

Infektiologie Kursus: Knochen- und Gelenkinfektionen

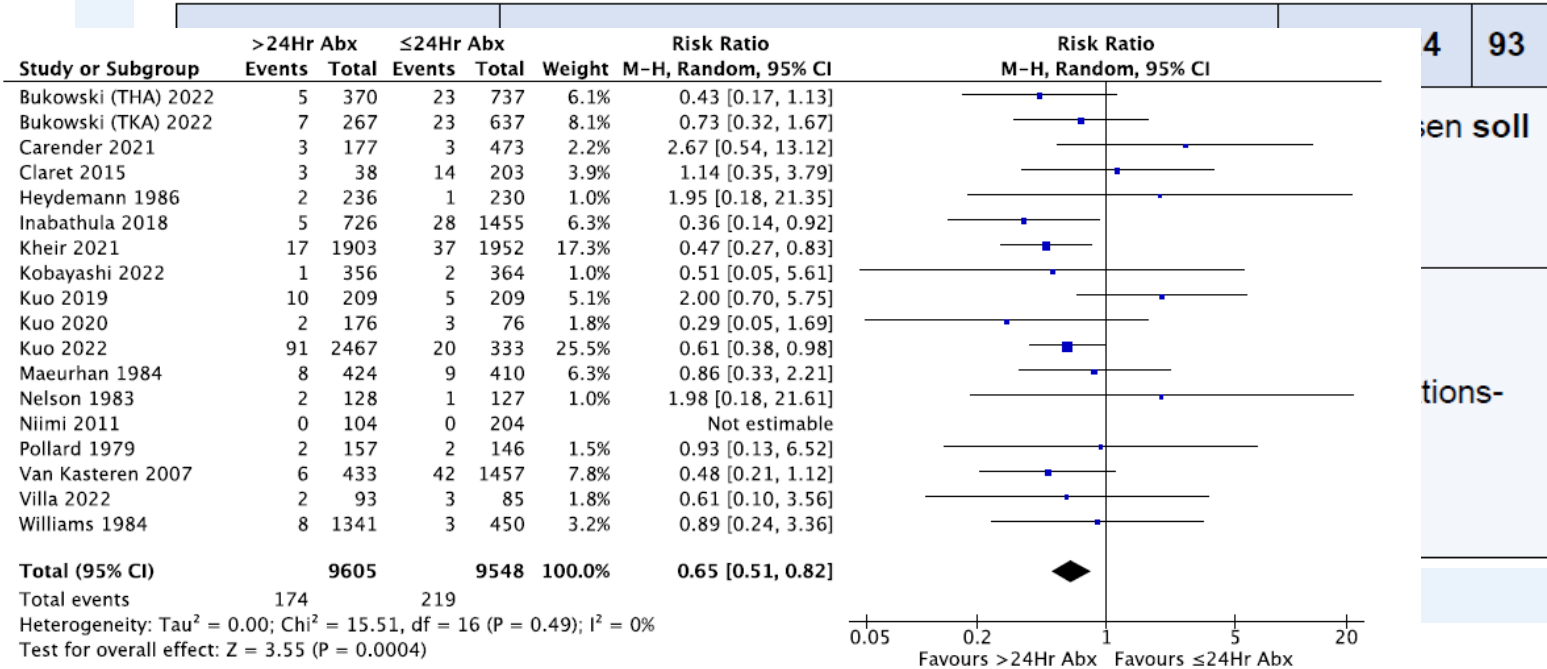
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Nächster Termin

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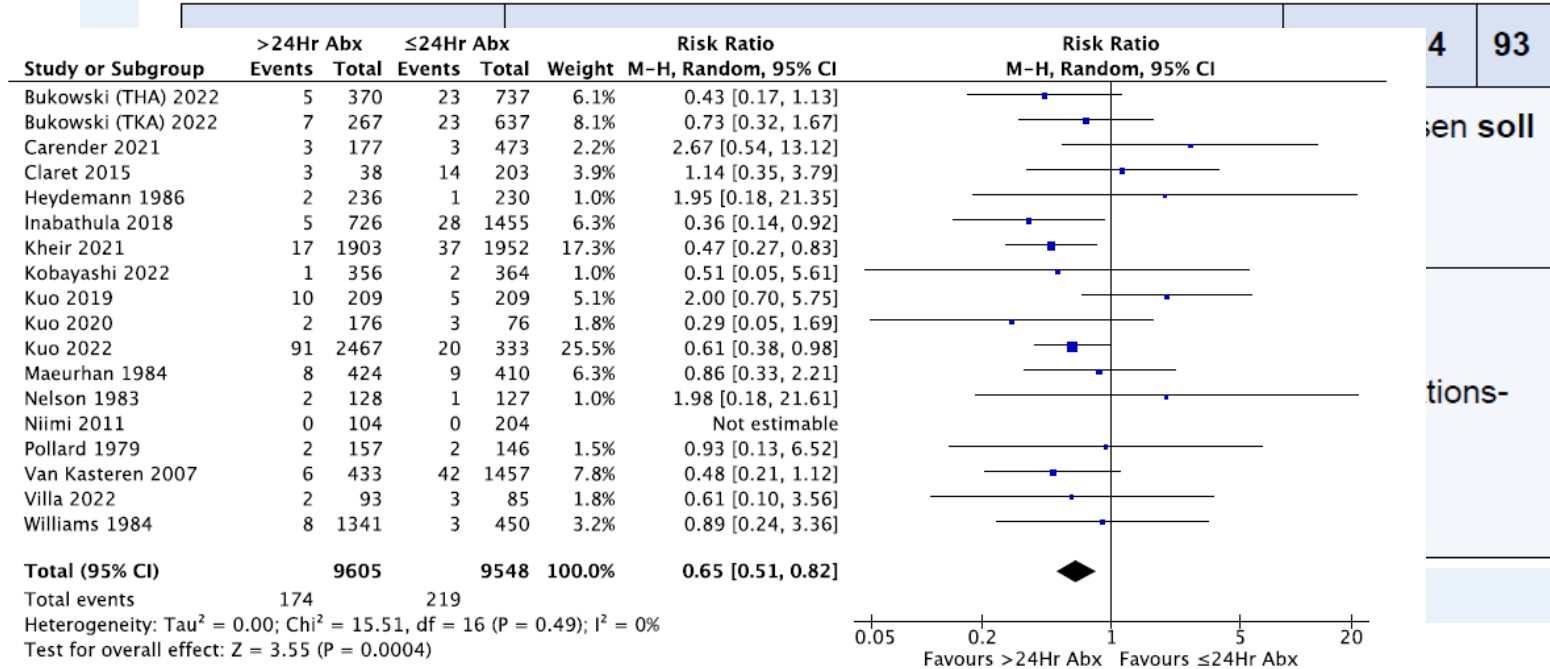
Online | **Online**

Conclusions: This meta-analysis demonstrated that patients treated with EAP were less likely to develop a PJI relative to those treated with the SoC for all TJA, primary TJA, primary THA, aseptic revision TJA, and aseptic revision TKA. There was no significant difference observed between EAP and SoC for primary TKA or aseptic revision THA.



Extended Prophylactic Antibiotics for Primary and Aseptic Revision Total Joint Arthroplasty: A Meta-Analysis

Conclusions: This meta-analysis demonstrated that patients treated with EAP were less likely to develop a PJI relative to those treated with the SoC for all TJA, primary TJA, primary THA, aseptic revision TJA, and aseptic revision TKA. There was no significant difference observed between EAP and SoC for primary TKA or aseptic revision THA.



Extended Prophylactic Antibiotics for Primary and Aseptic Revision Total Joint Arthroplasty: A Meta-Analysis

Gram-neg. zunehmend! Chinolonresistenz!



Internati

International Journal of Infectious Diseases 154 (2025) 107843

Contents lists available at ScienceDirect

Table 3

Rates of centre-specific microbiologic results over a specific time periods.

Episode / centre (n/N, %)					2016-2019 (n = 377)	P value
GPB	Zurich (n = 589)				210/242 (87)	0.22
	Geneva (n = 323)				58/67 (87)	0.83
	Las Palmas (n = 351)				-	0.03
	BCN (n = 86)				43/68 (63)	0.11
MSSA	Zurich (n = 589)				84/242 (35)	0.03
	Geneva (n = 323)				32/67 (48)	0.41
	Las Palmas (n = 351)				-	0.21
	BCN (n = 86)				17/68 (25)	0.48
MRSA ^a	Zurich (n = 204)				7/91 (8)	0.09
	Geneva (n = 181)				6/37 (16)	0.03
	Las Palmas (n = 211)				-	<0.01
	BCN (n = 36)				11/28 (39)	0.68
CoNS	Zurich (n = 589)				90/242 (37)	0.03
	Geneva (n = 323)	30/113 (27)	4/44 (9)	17/99 (17)	12/67 (18)	0.07
	Las Palmas (n = 351)	22/133 (17)	16/167 (10)	-	-	0.01
	BCN (n = 86)	-	-	4/18 (22)	4/68 (6)	0.03
GNB	Zurich (n = 589)	14/57 (25)	35/122 (29)	54/168 (32)	75/242 (31)	0.71
	Geneva (n = 323)	35/113 (31)	19/44 (43)	47/99 (47)	33/67 (49)	0.04
	Las Palmas (n = 351)	58/133 (43)	93/167 (56)	27/51 (53)	-	0.21
	BCN (n = 86)	-	-	9/18 (50)	47/68 (69)	0.13
Enterobacteriaceae	Zurich (n = 589)	12/57 (21)	15/122 (12)	38/168 (23)	58/242 (24)	0.07
	Geneva (n = 323)	18/113 (16)	13/44 (30)	33/99 (33)	26/67 (39)	<0.01
	Las Palmas (n = 351)	36/133 (27)	75/167 (45)	21/51 (41)	-	<0.01
	BCN (n = 86)	-	-	4/18 (22)	29/68 (43)	0.11

Gram-neg.:

Zürich: 25 -> 31%

Genf: 31 -> 49%

Las Palmas: 43 -> 53%

Barcelona : 50 -> 69%

Chinolon (R): 9 -> 14%



Posttraumatische Osteomyelitis – hohe Resistenzrate

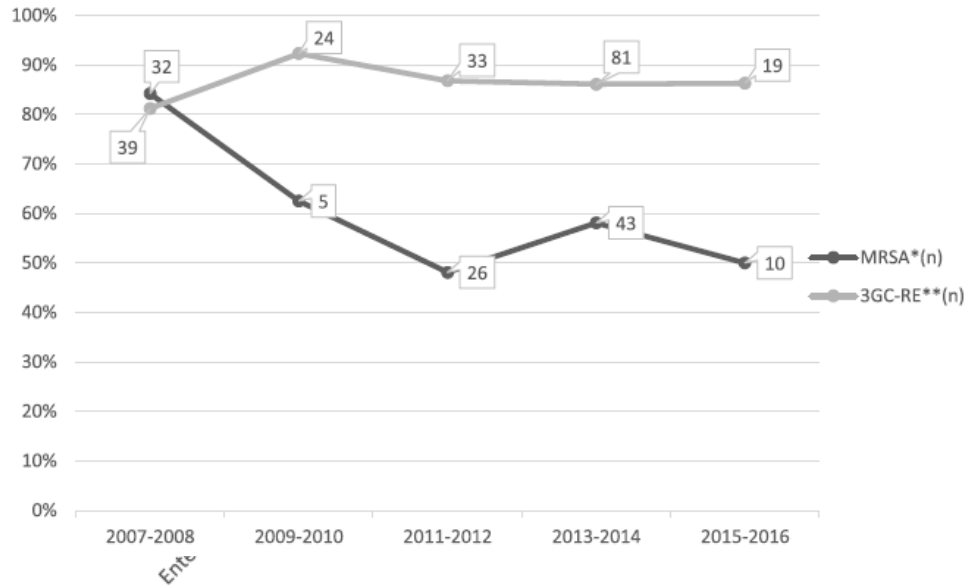


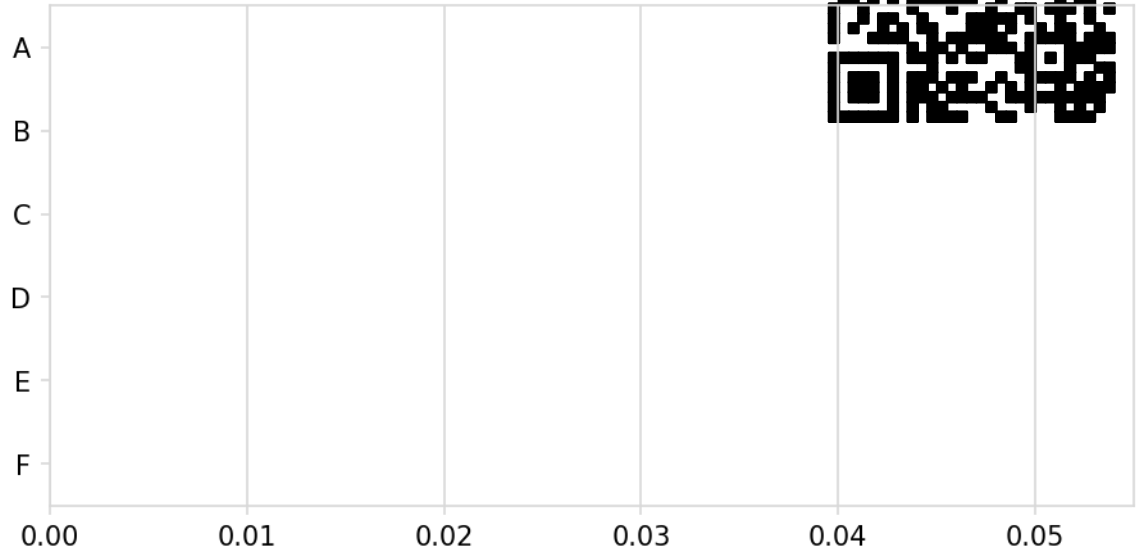
Fig. 1 Percentage of micro-organism among total isolates from bone biopsies in war-wounded civilians according to country of origin, 2006–2016.

**coagulase negative Staphylococcus*

	Enterobacteriaceae	
	N ^a	n (%)
Pip/tazo ^b	200	90 (45.0)
Cefepime	200	176 (88.0)
Imipenem	208	9 (4.3)
Ertapenem	85	8 (9.4)
Ciprofloxacin	200	117 (58.5)
Cotrimoxazole	179	150 (83.8)

In welchem Bereich arbeiten Sie?

- A) Chirurgie
- B) Innere/Allgemeinmedizin
- C) Anästhesie
- D) Pharmazie
- E) Mikrobiologie/Hygiene
- F) Andere Fachrichtung



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Umfrage noch nicht